

Nikita Tsoy

Sofia – Bulgaria

✉ nikita.tsoy@insait.ai • 🌐 nikita-tsoy98.github.io • in nikita-tsoy98

Doctoral researcher in trustworthy ML, with publications on incentives in ML systems, shortcut learning, federated learning under privacy concerns, and performative prediction. Background combines machine learning, economics, and mathematical modeling.

Skills

Machine Learning: deep learning, learning theory, shortcut learning, OOD generalization, differential privacy, federated learning, trustworthy ML

Statistics: econometrics, causal inference, quasi-experimental methods, time series

Mathematical Modeling: game theory, optimization, numerical methods

Programming: Python, PyTorch, NumPy, pandas, scikit-learn, Stata, R, C++

Languages: Russian (native), English

Research Experience

INSAIT, Sofia University

Doctoral Researcher

Advisor: Nikola Konstantinov

- Developed game-theoretic models of data-sharing incentives in ML systems, combining ML scaling laws with industrial organization models
- Analyzed participation incentives in federated learning under privacy concerns
- Studied shortcut learning in deep networks through theoretical analysis and controlled experiments on shortcut localization using a novel methodology
- Theoretically analyzed how model deployment changes the data distribution under performativity
- Contributed to empirical research on robustness of published econometric findings by implementing robustness checks and proposing weak-instrument diagnostics

ETH Zurich

Research Assistant in Machine Learning

- Investigated data sharing between competing firms, leading to a NeurIPS publication

Sofia, Bulgaria

2023–present

Zurich, Switzerland

2022–2023

Education

INSAIT, Sofia University

PhD in Machine Learning

New Economic School

MA in Economics, cum laude, GPA: 4.97/5 (top-1)

Moscow Institute of Physics and Technology

BS in Physics, cum laude, GPA: 4.91/5

Sofia, Bulgaria

2023–present

Moscow, Russia

2020–2022

Moscow, Russia

2016–2020

Publications

- 1: **Tsoy, N.**, Kirev, I., Rahimiyazdi, N., & Konstantinov, N. "On the Impact of Performative Risk Minimization for Binary Random Variables." *ICML*. 2025.
- 2: **Tsoy, N.**, & Konstantinov, N. "Simplicity Bias of Two-Layer Networks beyond Linearly Separable Data." *ICML*. 2024.
- 3: **Tsoy, N.**, Mihalkova, A., Todorova, T. N., & Konstantinov, N. "Provable Mutual Benefits from Federated Learning in Privacy-Sensitive Domains." *AISTATS*. 2024.
- 4: **Tsoy, N.**, & Konstantinov, N. "Strategic Data Sharing between Competitors." *NeurIPS*. 2023.

Working papers

- 1: **Tsoy, N.**, & Konstantinov, N. "On Measuring Localization of Shortcuts in Deep Networks." *SCSL Workshop*. 2025.
- 2: Chakarov, D., **Tsoy, N.**, Minchev, K., & Konstantinov, N. "Incentivizing Truthful Collaboration in Heterogeneous Federated Learning." *OPT Workshop*. 2024.
- 3: Campbell, D., Brodeur, A., Dreber, A., Johannesson, M., Kopecky, J., Lusher, L., & **Tsoy, N.** "The Robustness Reproducibility of the American Economic Review." *I4R Discussion Paper Series*. 2024.

Awards

- Petr Aven scholarship (top-3 students, 2021–2022)
- Abramov-Frolov Foundation Scholarship (top-10% students, 2016–2019)
- Bronze, "I am a Professional" Olympiad in Mathematics (2019)
- Awardee, Russian Olympiad in Mathematics (2015, 2016)
- First place, Belarusian Olympiad in Mathematics (2015)